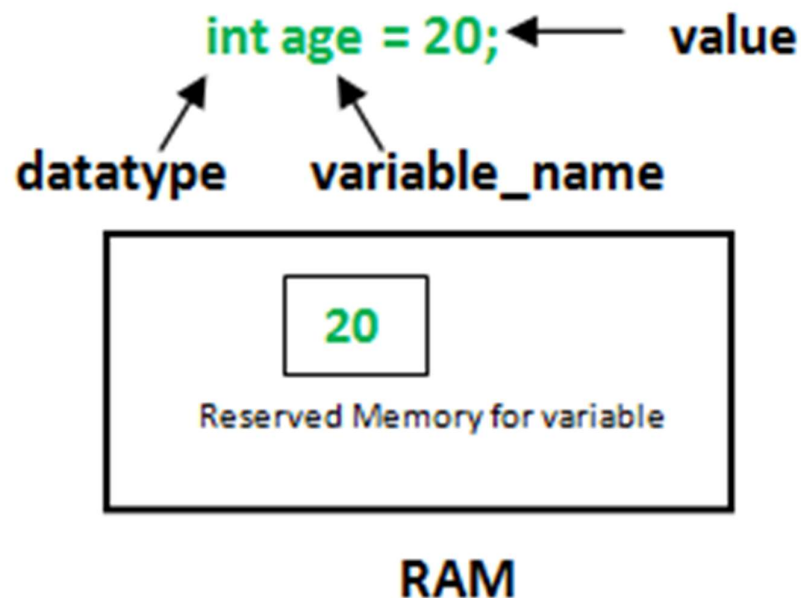


Variables in JAVA

What is Variable:-

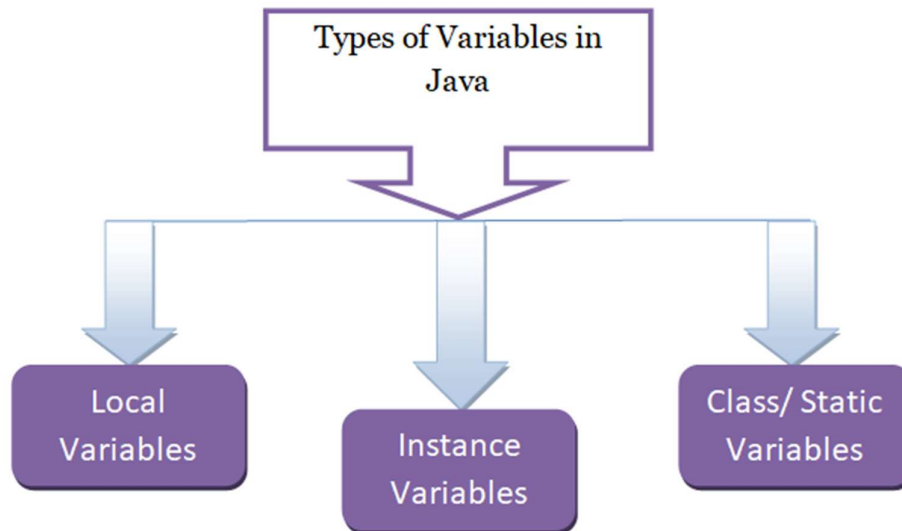
- In Java, variables are containers that **store data in memory**.
- Understanding variables plays a very important role as it defines how **data is stored, accessed, and manipulated**.



Key Components of Variables in Java:-

- **Data Type:** Defines the kind of data stored (e.g., int, String, float).
- **Variable Name:** A unique identifier following Java naming rules.
- **Value:** The actual data assigned to the variable.

✚ Types of Variable:-



✚ Local Variable :-

- Declared **inside a method, constructor, or block**.
- Created when the method is called and destroyed after the method ends.
- **No default value** (must be initialized).

Syntax:-

```
void display()  
{  
    int age = 20;           // local variable  
    System.out.println(age);  
}
```

Instance Variable :-

- Declared **inside a class** but **outside methods**.
- Each object gets its **own copy**.
- Has a default value if **not initialized**.

Syntax :-

```
class Example
{
    int age = 25;           // instance variable
}
```

Static Variable :-

- Declared with the **static keyword inside a class**.
- Shared by **all objects** of the class.
- Has a default **value if not initialized**.

Syntax :-

```
class Example
{
    static String company = "Fortune Cloud"; // static variable
}
```

Implementation

Input :-

```
1 package basicconcepts;
2
3 public class variabletypes {
4
5     int n=450;
6     static byte age=90;
7     static void first()
8     {
9         int num=60;
10        System.out.println("Value of number is : "+num);
11        System.out.println(age);
12    }
13    void second()
14    {
15        System.out.println("Value of number is" +n);
16    }
17    public static void main(String[] args)
18    {
19        System.out.println(age);
20    }
21 }
```

Output :-

90

Object Oriented Programming

What is OOP :-

- OOP is a programming paradigm based on the concept of objects, which contain data (**variables**) and behavior (**methods**).
- It helps to **organize code, improve reusability**, and make **programs modular and easier to maintain**.
- OOP in Java uses **classes and objects** to model real-world problems.
- It promotes **reusability, security, and scalability** using principles like **encapsulation, inheritance, polymorphism, and abstraction**.

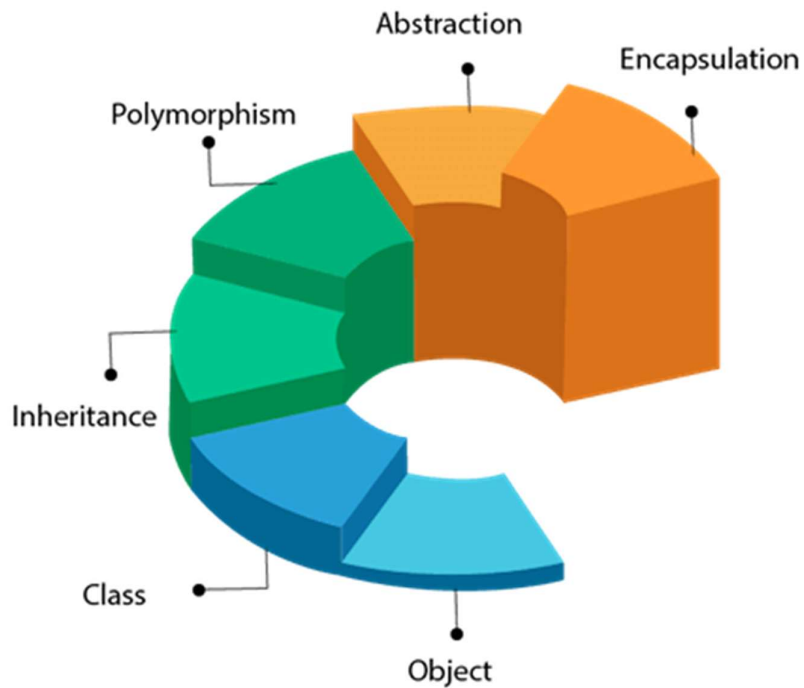
Advantages of OOP :-

- Code is **organized** into classes, making it easy to manage.
- **Reuse** code using inheritance.
- Easy to **add new features** without big changes.
- **Secure** by hiding data (encapsulation).
- Easy to **update and fix** programs.
- Shows only important details (abstraction).

Disadvantages of OOP :-

- **Planning and design** take more time.
- Runs a bit **slower** and uses more **memory**.
- Needs **more code** than simple programs.
- Some concepts are **hard for beginners** to understand.

OOPs (Object-Oriented Programming System)



Class :-

- A class is a **blueprint/template** to create objects.
- It defines **variables (data)** and **methods (behavior)**.

Object :-

- An object is a **real-world entity** created from a **class**.
- It represents an instance of the **class** and can **access variables and methods**.

Implementation

Input :-

```
1 package Oops;
2 public class person {
3     String name;
4     Long phone;
5
6     void walk()
7     {
8         System.out.println("Walking...");
9     }
10    void speak()
11    {
12        System.out.println("Speaking...");
13    }
14    public static void main(String[] args)
15    {
16        person p1=new person();
17        p1.name="Nikhil";
18        p1.phone=7949675931;
19
20        person p2=new person();
21        p2.name="Ram";
22        p2.phone=8868651341;
23
24        System.out.println("Person 1 Name : " +p1.name);
25        System.out.println("Person 1 Phone Number : " +p1.phone);
26
27        System.out.println("-----");
28        System.out.println("Person 2 name : " +p2.name);
29        System.out.println("Person 2 Phone Number : " +p2.phone);
30    }
31 }
```

Output :-

```
Person 1 Name : Nikhil
Person 1 Phone Number : 794967593
-----
Person 2 name : Ram
Person 2 Phone Number : 886865134
```